

$$\begin{aligned} & \frac{1}{16} m_2 \mu^2 g^4 (y_u^{i1i2} - \frac{1}{16} m_2 \mu^2 g^4 (y_u^{i1i2} - 2 y_u^{pi2} \delta_{pi1}) LF_{2,2,0}[m_2, \tilde{\mu}] + \\ & \frac{1}{6} m_2 \mu^2 g^4 (y_u^{i1i2} - 2 y_u^{pi2} \delta_{pi1}) LF_{3,2,-1}[m_2, \tilde{\mu}] - \\ & \frac{1}{4} \mu g_1^2 y_d^r y_u^{i1i2} \overline{a_d^p r} LF_{2,2,0}[m_d^r, m_q^p] - \frac{1}{4} \mu g_1^2 y_e^r y_u^{i1i2} \overline{a_e^p r} LF_{2,2,0}[m_e^r, m_l^p] + \\ & \frac{1}{2} \mu g_2^2 y_d^r y_u^{i1i2} \overline{a_e^p r} LF_{3,1,0}[m_l^p, m_e^r] + \frac{1}{4} \mu g_1^2 y_e^r y_u^{i1i2} \overline{a_e^p r} LF_{3,2,-1}[m_l^p, m_e^r] + \\ & \frac{1}{8} \mu y_d^r \overline{a_e^p r} (y_u^{i1i2} (3 g_1^2 - 11 g_2^2) + 2 g_2^2 y_u^{s12} \delta_{s11}) LF_{4,1,-1}[m_l^p, m_e^r] + \\ & \frac{1}{6} \mu y_e^r \overline{a_e^p r} (y_u^{i1i2} (-3 g_1^2 + 4 g_2^2) - 2 g_2^2 y_u^{s12} \delta_{s11}) LF_{5,1,-2}[m_l^p, m_e^r] + \frac{1}{2} \mu y_d^r y_u^{i1i2} \\ & \overline{a_d^p r} (-2 g_1^2 + 3 g_2^2) LF_{3,1,0}[m_q^p, m_d^r] + \frac{1}{4} \mu g_1^2 y_d^r y_u^{i1i2} \overline{a_d^p r} LF_{3,2,-1}[m_q^p, m_d^r] + \\ & \frac{3}{8} \mu y_d^r \overline{a_d^p r} (y_u^{i1i2} (7 g_1^2 - 11 g_2^2) + 2 g_2^2 y_u^{s12} \delta_{s11}) LF_{4,1,-1}[m_q^p, m_d^r] + \\ & \frac{1}{2} \mu y_d^r \overline{a_d^p r} (y_u^{i1i2} (-3 g_1^2 + 4 g_2^2) - 2 g_2^2 y_u^{s12} \delta_{s11}) LF_{5,1,-2}[m_q^p, m_d^r] - \\ & \frac{1}{8} \mu y_d^r \overline{a_d^p r} (y_u^{i1i2} (g_1^2 + 5 g_2^2) - 4 g_2^2 y_u^{s12} \delta_{s11}) LF_{2,2,0}[m_q^p, m_d^r] + \\ & \frac{3}{4} \mu g_2^2 y_d^r y_u^{i1i2} \overline{a_d^p r} LF_{3,1,0}[m_q^p, m_d^r] - \frac{1}{4} \mu g_2^2 y_d^r \overline{a_d^p r} (y_u^{i1i2} + y_u^{s12} \delta_{s11}) \\ & LF_{3,2,-1}[m_q^p, m_d^r] - \frac{3}{4} \mu g_2^2 y_d^r y_u^{i1i2} \overline{a_d^p r} LF_{4,1,-1}[m_q^p, m_d^r] + \\ & \frac{1}{4} \mu y_d^r \overline{a_d^p r} (y_u^{i1i2} (g_1^2 + 4 g_2^2) - 2 g_2^2 y_u^{s12} \delta_{s11}) LF_{3,1,0}[m_u^r, m_q^p] + \\ & \frac{1}{8} \mu y_d^r \overline{a_d^p r} (y_u^{i1i2} (g_1^2 + 5 g_2^2) - 4 g_2^2 y_u^{s12} \delta_{s11}) LF_{3,2,-1}[m_u^r, m_q^p] + \\ & \frac{3}{4} \mu y_d^r \overline{a_d^p r} (y_u^{i1i2} (g_1^2 - 4 g_2^2) + 2 g_2^2 y_u^{s12} \delta_{s11}) LF_{4,1,-1}[m_u^r, m_q^p] + \\ & \frac{1}{2} \mu y_d^r \overline{a_d^p r} (y_u^{i1i2} (-3 g_1^2 + 4 g_2^2) - 2 g_2^2 y_u^{s12} \delta_{s11}) LF_{5,1,-2}[m_u^r, m_q^p] + \\ & \frac{1}{4} m_1 \mu g_1^2 y_u^{i1i2} (-g_1^2 + g_2^2) LF_{3,1,0}[\tilde{\mu}, m_1] - \frac{1}{2} m_1 \mu g_1^4 y_u^{i1i2} LF_{3,2,-1}[\tilde{\mu}, m_1] + \\ & \frac{1}{8} m_1 \mu g_1^2 (y_u^{i1i2} (6 g_1^2 - 7 g_2^2) + 2 g_2^2 y_u^{pi2} \delta_{pi1}) LF_{4,1,-1}[\tilde{\mu}, m_1] + \\ & \frac{1}{2} m_1 \mu g_1^4 y_u^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6} m_1 \mu g_1^2 (y_u^{i1i2} (-3 g_1^2 + 4 g_2^2) - 2 g_2^2 y_u^{pi2} \delta_{pi1}) \\ & LF_{5,1,-2}[\tilde{\mu}, m_1] + \frac{1}{12} m_2 \mu \tilde{\mu} g_2^2 (y_u^{i1i2} (-9 g_1^2 + 13 g_2^2) - 8 g_2^2 y_u^{pi2} \delta_{pi1}) LF_{3,1,0}[\tilde{\mu}, m_2] - \\ & \frac{1}{3} m_2 \mu \tilde{\mu} g_2^4 (5 y_u^{i1i2} + 2 y_u^{pi2} \delta_{pi1}) LF_{3,2,-1}[\tilde{\mu}, m_2] + \frac{1}{8} m_2 \mu \tilde{\mu} g_2^2 \\ & (y_u^{i1i2} (18 g_1^2 - 25 g_2^2) + 14 g_2^2 y_u^{pi2} \delta_{pi1}) LF_{4,1,-1}[\tilde{\mu}, m_2] + 2 m_2 \mu \tilde{\mu} g_2^4 y_u^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_2] \\ & \frac{1}{2} m_2 \mu \tilde{\mu} g_2^2 (y_u^{i1i2} (-3 g_1^2 + 4 g_2^2) - 2 g_2^2 y_u^{pi2} \delta_{pi1}) LF_{5,1,-2}[\tilde{\mu}, m_2] + \\ & \frac{1}{36} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} LF_{2,1,1,0}[m_1, m_d^r, m_q^{i1}] + \frac{1}{72} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} \\ & LF_{2,2,1,-1}[m_1, m_d^r, m_q^{i1}] - \frac{1}{36} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} LF_{3,1,1,-1}[m_1, m_d^r, m_q^{i1}] + \\ & \frac{1}{6} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} LF_{2,1,1,0}[m_1, m_d^r, \tilde{\mu}] - \frac{1}{6} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} \\ & LF_{3,1,1,-1}[m_1, m_d^r, \tilde{\mu}] - \frac{1}{72} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} LF_{2,2,1,-1}[m_1, m_q^{i1}, m_d^r] + \\ & \frac{1}{18} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} LF_{2,1,1,0}[m_1, m_q^{i1}, m_u^r] - \frac{1}{36} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} y_u^{i1r} \\ & LF_{2,2,1,-1}[m_1, m_q^{i1}, m_u^r] - \frac{1}{18} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} y_u^{i1r} LF_{3,1,1,-1}[m_1, m_q^{i1}, m_u^r] + \\ & \frac{1}{72} m_1 \mu \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 - g_2^2) LF_{2,2,1,-1}[m_1, m_q^{i1}, m_u^{i2}] + \frac{1}{36} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} \\ & y_u^{i1r} LF_{2,2,1,-1}[m_1, m_u^r, m_q^{i1}] - \frac{1}{3} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} y_u^{i1r} LF_{2,1,1,0}[m_1, m_u^r, \tilde{\mu}] + \\ & \frac{1}{3} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} y_u^{i1r} LF_{3,1,1,-1}[m_1, m_u^r, \tilde{\mu}] + \frac{1}{72} m_1 \mu \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 - g_2^2) \\ & LF_{2,2,1,-1}[m_1, m_u^{i2}, m_q^{i1}] - \frac{5}{24} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_d^r] + \\ & \frac{1}{48} m_1 \mu \tilde{\mu} g_1^2 (-2 y_u^{pi2} (\overline{y_d^p r} y_d^{i1r} + \overline{y_d^p r} y_u^{i1r}) + y_u^{i1i2} (-9 g_1^2 + g_2^2)) LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_q^{i1}] + \\ & \frac{7}{24} m_1 \mu \tilde{\mu} g_1^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} y_u^{i1r} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_u^r] + \\ & \frac{1}{12} m_1 \mu \tilde{\mu} g_1^2 y_u^{i1i2} (9 g_1^2 - g_2^2) LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_u^{i2}] - \frac{3}{4} \mu g_1^2 g_2^2 y_u^{i1i2} \\ & (3 m_1 + m_2) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \mu g_1^2 g_2^2 y_u^{i1i2} (m_1 + m_2) LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\ & \frac{1}{8} m_2 \mu \tilde{\mu} g_2^2 \overline{y_d^p r} y_d^{i1r} y_u^{pi2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_d^r] - \frac{1}{2} m_2 \mu \tilde{\mu} g_2^4 y_u^{pi2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^p] \delta_{pi1} \\ & \frac{1}{16} m_2 \mu \tilde{\mu} g_2^2 (6 y_u^{pi2} (-\overline{y_d^p r} y_d^{i1r} + \overline{y_d^p r} y_u^{i1r}) + y_u^{i1i2} (3 g_1^2 + 29 g_2^2)) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^{i1}] \\ & \frac{1}{8} m_2 \$$